Basic Trouble shooting commands ipconfig, ping , tracert , nslookup, arp

Aim: To study and execute basic network troubleshooting commands such as ipconfig, ping, tracert, nslookup, arp, and netstat in Windows to diagnose network connectivity issues, examine IP configuration, trace network routes, resolve domain names, and inspect network connections.

Description:

Network troubleshooting commands help identify and solve problems related to IP addresses, network connectivity, routing, DNS, and MAC addresses. These commands are commonly executed in Command Prompt (CMD) on Windows. To open Command Prompt, press Win + R, type cmd, and press Enter.

1. `ipconfig` — IP Configuration

Purpose: Displays the IP configuration of a computer, including its IP address, subnet mask, and default gateway.

1. Commands:
2. ipconfig
3. ipconfig /all
4. ipconfig /release
5. ipconfig /renew
6. ipconfig /flushdns
7. `ipconfig /displaydns`

- `ipconfig` — Displays basic IP configuration.
- `ipconfig /all` — Displays detailed network adapter information, including MAC address and DNS servers.
- `ipconfig /release` — Releases the IPv4 address obtained through DHCP.
- `ipconfig /renew` — Requests a new IPv4 address through DHCP.
- `ipconfig /flushdns` — Clears the local DNS resolver cache.

Example output:

IPv4 Address . . . . . : 192.168.1.10

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . : 192.168.1.1

2. `ping` — Test Network Connectivity

Purpose: Checks whether a destination device is reachable over an IP network and measures round-trip time.

Command: `ping 192.168.1.1`

To test internet connectivity:

```
ping 8.8.8.8
```

Example output:

```
Reply from 192.168.1.1:
bytes=32 time=2ms TTL=64
```

Interpretation:

- Reply received: The destination responded to the ping.
- Request timed out: No response arrived within the timeout period.
- Destination host unreachable: A device or router reports that it cannot reach the destination.

Note: A failed ping does not always mean the device is offline; firewalls may block ICMP traffic.

3. `tracert` — Trace the Route

Purpose: `tracert` stands for Trace Route. It is a Windows command used to identify the routers (hops) that packets pass through when travelling from your computer to a destination, such as a website.

It helps you understand the path taken by network traffic and troubleshoot connectivity problems.

Command: `tracert google.com`

Example output:

```
Tracing route to google.com
1    2 ms    1 ms    2 ms    192.168.1.1
2   10 ms   12 ms   11 ms    10.10.0.1
3   25 ms   24 ms   26 ms    ...
```

Explanation:

- Each numbered line represents a hop, usually a router.
- The three time values represent round-trip time measurements for three probes.
- `* * *` indicates that a probe did not receive a response within the timeout period.

Use: Helps identify where along a network route delays or connectivity problems may occur.

4. `nslookup` — DNS Lookup

Purpose: The `nslookup` (Name Server Lookup) command is used to find the IP address of a domain name using DNS (Domain Name System).

Syntax : nslookup Domain_name [optional DNS Server Ip address]

Parameter	Description
`Nslookup`	Name Server Lookup command
`domain_name`	The domain name whose IP address we want to find
`[dns_address]`	IP address of the DNS server to which the query is sent

Example 1:

C:\Users\MGIT>nslookup www.mgit.ac.in

Server: ns3.dvpl.in
Address: 119.235.51.170

Non-authoritative answer:
Name: www.mgit.ac.in
Address: 65.2.53.139

Output Explaination:

- Server: `ns3.dvpl.in` is the DNS server that your computer contacted.
- Address: `119.235.51.170` is the IP address of that DNS server.
- Non-authoritative answer: The response came from a DNS server that obtained the information through caching or another DNS lookup, rather than directly from the domain's authoritative DNS server.
- Name: `www.mgit.ac.in` is the domain name you queried.
- Address: `65.2.53.139` is the IP address associated with that domain name.

Example 2:

Command: `nslookup google.com`

```
Server:  dns.example.com
Address: 192.168.1.1

Non-authoritative answer:
Name:    google.com
Addresses: 142.250.x.x
```

Explanation:

- Displays the DNS server used for the query.
- Returns IP address records for the requested domain.
- Helps identify DNS resolution problems.

To query a specific DNS server:

`Example 3:`

Command: nslookup google.com 8.8.8.8

- `www.google.com` → Domain name to be resolved.
- `8.8.8.8` → Google Public DNS server.
- The DNS server returns the IP address associated with `www.google.com`

5. arp — Address Resolution Protocol

Purpose: ARP (Address Resolution Protocol) is used to find the MAC address corresponding to an IP address in a local area network (LAN).

Commands:

```
arp -a
arp -d *
arp -s 192.168.1.20 00-AA-BB-CC-DD-EE
```

Explanation:

- `arp -a` — Displays cached IP-to-MAC address mappings.
- `arp -d *` — Attempts to delete dynamic ARP cache entries.
- `arp -s` — Adds a static ARP entry.

Example output:

```
C:\Users\MGIT>arp -a

      Interface: 172.24.106.118 --- 0x4

        Internet Address      Physical Address      Type

        172.24.8.4            34-9f-7b-b6-07-2c     dynamic

        172.24.8.8            7c-4d-8f-84-4c-dd     dynamic

        172.24.8.56           00-17-c8-52-07-9e     dynamic

        172.24.8.57           74-38-b7-52-58-b6     dynamic
```

Use: Helps troubleshoot local network communication and IP-to-MAC address resolution.

Other commands are

1. `Netstat` : Displays connections and ports
2. `route print` : Displays routing tables
3. `pathping` : Analyzes route delays and packet loss
4. `hostname`: Displays the computer name